KERALA DIPLOMA CURRICULUM REVISION 2026

Course 5493B | 5 Credits: 4 | Program Elective | Source-grounded

Digital CMOS VLSI

- To learn the fundamentals of VLSI design

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 5493B is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5493B.pdf from the uploaded official SITTTR syllabus archive.
Course code and title: preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Microelectronics
Course code	5493B
Course title	Digital CMOS VLSI
Semester	5 Credits: 4
Course category	Program Elective
Periods per week	4 (L:4, T:0, P:0) Periods per semester: 60
Periods per semester	60
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

16 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- ● To learn the fundamentals of VLSI design
- ● To familiarize the implementation of static and dynamic CMOS logic design.
- ● To gain knowledge in arithmetic building blocks and memory architectures.

Course outcomes

CO	Official outcome	Study level
CO1	16 Understanding FPGA design. Design VLSI Logic circuits with various MOSFET	See official syllabus
CO2	12 Applying logic families.	See official syllabus
CO3	Compare different types of memory elements. 14 Understanding Design and analyse data path elements such as	See official syllabus
CO4	16 Applying Adders and multipliers.	See official syllabus

CO–PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3
CO2	CO2 3
CO3	CO3 3
CO4	CO4 3

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	Explain the various methodologies in ASIC and FPGA design.	4	As specified
Module 2	Design VLSI Logic circuits with various MOSFET logic families.	4	As specified
Module 3	Compare different types of memory elements.	4	As specified
Module 4	Design and analyse data path elements such as Adders and multipliers.	4	As specified

MODULE 1

Explain the various methodologies in ASIC and FPGA design.

This module is presented from the official syllabus scope for Digital CMOS VLSI. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Explain the various methodologies in ASIC and FPGA design.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▼ 3 Understanding ASICs

3 Understanding ASICs is an official topic in Module 1 of Digital CMOS VLSI. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding ASICs using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding asics should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding ASICs means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ 3 Understanding ASICs □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▼ Explain the processes involved in SoC design.

Explain the processes involved in SoC design. is an official topic in Module 1 of Digital CMOS VLSI. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the processes involved in SoC design. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the processes involved in soc design. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain the processes involved in SoC design. means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Explain the processes involved in SoC design. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▼ Outline the FPGA design flow 4 Understanding Compare the speed, power and area

Outline the FPGA design flow 4 Understanding Compare the speed, power and area is an official topic in Module 1 of Digital CMOS VLSI. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Outline the FPGA design flow 4 Understanding Compare the speed, power and area using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, outline the fpga design flow 4 understanding compare the speed, power and area should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare

Aspect	What to prepare
Definition/identity	What Outline the FPGA design flow 4 Understanding Compare the speed, power and area means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Outline the FPGA design flow 4 Understanding Compare the speed, power and area □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▼ 6 Understanding considerations in VLSI VLSI Design Methodologies. Introduction: Moore’s law .ASIC design, Full custom ASICs, Standard cell based ASICs, Gate array based ASICs, SoCs, FPGA devices, FPGA Design flows, Top-Down and Bottom-Up design methodologies. Speed power and area considerations in VLSI design

6 Understanding considerations in VLSI VLSI Design Methodologies. Introduction: Moore’s law .ASIC design, Full custom ASICs, Standard cell based ASICs, Gate array based ASICs, SoCs, FPGA devices, FPGA Design flows, Top-Down and Bottom-Up design methodologies. Speed power and area considerations in VLSI design is an official topic in Module 1 of Digital CMOS VLSI. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 6 Understanding considerations in VLSI VLSI Design Methodologies. Introduction: Moore’s law .ASIC design, Full custom ASICs, Standard cell based ASICs, Gate array based ASICs, SoCs, FPGA devices, FPGA Design flows, Top-Down and Bottom-Up design methodologies. Speed power and area considerations in VLSI design using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 6 understanding considerations in vlsi vlsi design methodologies. introduction: moore’s law .asic design, full custom asics, standard cell based asics, gate array based asics, socs, fpga devices, fpga design flows, top-down and bottom-up design methodologies. speed power and area considerations in vlsi design should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 6 Understanding considerations in VLSI VLSI Design Methodologies. Introduction: Moore’s law .ASIC design, Full custom ASICs, Standard cell based ASICs, Gate array based ASICs, SoCs, FPGA devices, FPGA Design flows, Top-Down and Bottom-Up design methodologies. Speed power and area considerations in VLSI design means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ 6 Understanding considerations in VLSI VLSI Design Methodologies. Introduction: Moore’s law .ASIC design, Full custom ASICs, Standard cell based ASICs, Gate array based ASICs, SoCs, FPGA devices, FPGA Design flows, Top-Down and Bottom-Up design methodologies. Speed power and area considerations in VLSI design [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

▶ **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

Design VLSI Logic circuits with various MOSFET logic families.

This module is presented from the official syllabus scope for Digital CMOS VLSI. Use the topic cards for learning and the source transcript for exact coverage.

As specified	**CO-linked**
official hours	see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Design VLSI Logic circuits with various MOSFET logic families.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▼ 3 Understanding inverter Realize the given logic function using static

3 Understanding inverter Realize the given logic function using static is an official topic in Module 2 of Digital CMOS VLSI. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding inverter Realize the given logic function using static using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding inverter realize the given logic function using static should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare

Aspect	What to prepare
Definition/identity	What 3 Understanding inverter Realize the given logic function using static means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 3 Understanding inverter Realize the given logic function using static □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▼ 3 Applying CMOS logic and transmission gate logic. Compare and contrast static and dynamic

3 Applying CMOS logic and transmission gate logic. Compare and contrast static and dynamic is an official topic in Module 2 of Digital CMOS VLSI. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Applying CMOS logic and transmission gate logic. Compare and contrast static and dynamic using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 applying cmos logic and transmission gate logic. compare and contrast static and dynamic should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare

Aspect	What to prepare
Definition/identity	What 3 Applying CMOS logic and transmission gate logic. Compare and contrast static and dynamic means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 3 Applying CMOS logic and transmission gate logic. Compare and contrast static and dynamic □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▾ 2 Understanding CMOS logic. Realize the given logic function using pass

2 Understanding CMOS logic. Realize the given logic function using pass is an official topic in Module 2 of Digital CMOS VLSI. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding CMOS logic. Realize the given logic function using pass using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding cmos logic. realize the given logic function using pass should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare

Aspect	What to prepare
Definition/identity	What 2 Understanding CMOS logic. Realize the given logic function using pass means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 2 Understanding CMOS logic. Realize the given logic function using pass □□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

▼ 4 Applying transistor logic and transmission gate logic. Static CMOS Logic Design MOSFET Logic Design - NMOS Inverter (Static analysis only), basic logic gates, CMOS logic, Static and transient analysis of CMOS inverter. Realization of logic functions with static CMOS logic, Pass transistor logic, and transmission gate logic

4 Applying transistor logic and transmission gate logic. Static CMOS Logic Design MOSFET Logic Design - NMOS Inverter (Static analysis only), basic logic gates, CMOS logic, Static and transient analysis of CMOS inverter. Realization of logic functions with static CMOS logic, Pass transistor logic, and transmission gate logic is an official topic in Module 2 of Digital CMOS VLSI. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Applying transistor logic and transmission gate logic. Static CMOS Logic Design MOSFET Logic Design - NMOS Inverter (Static analysis only), basic logic gates, CMOS logic, Static and transient analysis of CMOS inverter. Realization of logic functions with static CMOS logic, Pass transistor logic, and transmission gate logic using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 applying transistor logic and transmission gate logic. static cmos logic design mosfet logic design - nmos inverter (static analysis only), basic logic gates, cmos logic, static and transient analysis of cmos inverter. realization of logic functions with static cmos logic, pass transistor logic, and transmission gate logic should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Applying transistor logic and transmission gate logic. Static CMOS Logic Design MOSFET Logic Design - NMOS Inverter (Static analysis only), basic logic gates, CMOS logic, Static and transient analysis of CMOS inverter. Realization of logic functions with static CMOS logic, Pass transistor logic, and transmission gate logic means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 4 Applying transistor logic and transmission gate logic. Static CMOS Logic Design MOSFET Logic Design - NMOS Inverter (Static analysis only), basic logic gates, CMOS logic, Static and transient analysis of CMOS inverter. Realization of logic functions with static CMOS logic, Pass transistor logic, and transmission gate logic [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

▶ **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Compare different types of memory elements.

This module is presented from the official syllabus scope for Digital CMOS VLSI. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Compare different types of memory elements.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▼ 3 Understanding and its drawbacks.

3 Understanding and its drawbacks. is an official topic in Module 3 of Digital CMOS VLSI. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding and its drawbacks. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding and its drawbacks. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding and its drawbacks. means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ 3 Understanding and its drawbacks. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▼ Explain NP domino logic.

Explain NP domino logic. is an official topic in Module 3 of Digital CMOS VLSI. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain NP domino logic. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain np domino logic. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain NP domino logic. means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Explain NP domino logic. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▼ Realize OR, NOR and NAND ROM cells.

Realize OR, NOR and NAND ROM cells. is an official topic in Module 3 of Digital CMOS VLSI. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Realize OR, NOR and NAND ROM cells. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, realize or, nor and nand rom cells. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Realize OR, NOR and NAND ROM cells. means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Realize OR, NOR and NAND ROM cells. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▼ **Compare and contrast SRAM and DRAM 2 Understanding Dynamic logic Design and Storage Cells Dynamic Logic Design-Pre charge- Evaluate logic, Domino Logic, NP domino logic. Read Only Memory-4x4 MOS ROM Cell Arrays(OR,NOR,NAND) Random Access Memory –SRAM-Six transistor CMOS SRAM cell, DRAM –Three transistor and One transistor Dynamic Memory Cell..**

Compare and contrast SRAM and DRAM 2 Understanding Dynamic logic Design and Storage Cells Dynamic Logic Design-Pre charge- Evaluate logic, Domino Logic, NP domino logic. Read Only Memory-4x4 MOS ROM Cell Arrays(OR,NOR,NAND) Random Access Memory –SRAM-Six transistor CMOS SRAM cell, DRAM –Three transistor and One transistor Dynamic Memory Cell.. is an official topic in Module 3 of Digital CMOS VLSI. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Compare and contrast SRAM and DRAM 2 Understanding Dynamic logic Design and Storage Cells Dynamic Logic Design-Pre charge- Evaluate logic, Domino Logic, NP domino logic. Read Only Memory-4x4 MOS ROM Cell Arrays(OR,NOR,NAND) Random Access Memory –SRAM-Six transistor CMOS SRAM cell, DRAM –Three transistor and One transistor Dynamic Memory Cell.. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.

- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, compare and contrast sram and dram 2 understanding dynamic logic design and storage cells dynamic logic design-pre charge-evaluate logic, domino logic, np domino logic. read only memory-4x4 mos rom cell arrays(or,nor,nand) random access memory –sram-six transistor cmos sram cell, dram –three transistor and one transistor dynamic memory cell.. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Compare and contrast SRAM and DRAM 2 Understanding Dynamic logic Design and Storage Cells Dynamic Logic Design-Pre charge-Evaluate logic, Domino Logic, NP domino logic. Read Only Memory-4x4 MOS ROM Cell Arrays(OR,NOR,NAND) Random Access Memory –SRAM-Six transistor CMOS SRAM cell, DRAM –Three transistor and One transistor Dynamic Memory Cell.. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Compare and contrast SRAM and DRAM 2 Understanding Dynamic logic Design and Storage Cells Dynamic Logic Design-Pre charge- Evaluate logic, Domino Logic, NP domino logic. Read

Only Memory-4x4 MOS ROM Cell Arrays(OR,NOR,NAND) Random Access Memory –SRAM-Six transistor CMOS SRAM cell, DRAM –Three transistor and One transistor Dynamic Memory Cell.. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▶ **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

Design and analyse data path elements such as Adders and multipliers.

This module is presented from the official syllabus scope for Digital CMOS VLSI. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Design and analyse data path elements such as Adders and multipliers.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▼ 2 Understanding bypass adder.

2 Understanding bypass adder. is an official topic in Module 4 of Digital CMOS VLSI. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding bypass adder. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding bypass adder. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding bypass adder. means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ 2 Understanding bypass adder. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▼ Explain Linear carry select adder working. 5 Understanding Explain the working of square root carry

Explain Linear carry select adder working. 5 Understanding Explain the working of square root carry is an official topic in Module 4 of Digital CMOS VLSI. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain Linear carry select adder working. 5 Understanding Explain the working of square root carry using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain linear carry select adder working. 5 understanding explain the working of square root carry should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare

Aspect	What to prepare
Definition/identity	What Explain Linear carry select adder working. 5 Understanding Explain the working of square root carry means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Explain Linear carry select adder working. 5 Understanding Explain the working of square root carry □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□□, steps, application □□□□□□ □□□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

▼ 5 Understanding select adder.

5 Understanding select adder. is an official topic in Module 4 of Digital CMOS VLSI. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 5 Understanding select adder. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 5 understanding select adder. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 5 Understanding select adder. means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

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Malayalam support: Module 4-□□ 5 Understanding select adder. □□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

▼ **Explain an array multiplier. 4 Understanding Arithmetic circuits Adders: Static adder, Carry-Bypass adder, Linear Carry- Select adder, Square- root carry- select adder. Multipliers: Array multiplier. Text /Reference: T/R BookTitle/Author Michael John Sebastian Smith, Application Specific Integrated Circuits, Pearson T1 Education,2001. Sung –Mo Kang & Yusuf Leblebici, CMOS Digital Integrated Circuits- Analysis T2 & Design, McGraw-Hill, Third Ed., 2003. Jan M. Rabaey, Digital Integrated Circuits- A Design Perspective, Prentice Hall, R3 Second Edition, 2005. R4 Modern VLSI Design – Wayne Wolf, 3 Ed., 1997, Pearson Education. CMOS VLSI Design-A Circuits and Systems Perspective, Neil H.E Weste, R5 David Harris, Ayan Banerjee, 3rd Edn, Pearson, 2009. Online Resources: Sl. No Website Link 1 https://onlinecourses.nptel.ac.in/noc24_ee29/preview 2 https://pages.hmc.edu/harris/cmosvlsi/4e/index.html 3 http://www.vlsi-expert.com/ 4 https://semiwiki.com/**

Explain an array multiplier. 4 Understanding Arithmetic circuits Adders: Static adder, Carry-Bypass adder, Linear Carry- Select adder, Square- root carry- select adder. Multipliers: Array multiplier. Text /Reference: T/R BookTitle/Author Michael John Sebastian Smith, Application Specific Integrated Circuits, Pearson T1 Education,2001. Sung –Mo Kang & Yusuf Leblebici, CMOS Digital Integrated Circuits- Analysis T2 & Design, McGraw-Hill, Third Ed., 2003. Jan M. Rabaey, Digital Integrated Circuits- A Design Perspective, Prentice Hall, R3 Second Edition, 2005. R4 Modern VLSI Design – Wayne Wolf, 3 Ed., 1997, Pearson Education. CMOS VLSI Design-A Circuits and Systems Perspective, Neil H.E Weste, R5 David Harris, Ayan Banerjee, 3rd Edn, Pearson, 2009. Online Resources: Sl. No Website Link 1 https://onlinecourses.nptel.ac.in/noc24_ee29/preview 2 https://pages.hmc.edu/harris/cmosvlsi/4e/index.html 3 http://www.vlsi-expert.com/ 4 https://semiwiki.com/ is an official topic in Module 4 of Digital CMOS VLSI. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain an array multiplier. 4 Understanding Arithmetic circuits Adders: Static adder, Carry-Bypass adder, Linear Carry- Select adder, Square- root carry- select adder. Multipliers: Array multiplier. Text /Reference: T/R BookTitle/Author Michael John Sebastian Smith, Application Specific Integrated Circuits, Pearson T1 Education,2001. Sung –Mo Kang & Yusuf Leblebici, CMOS Digital Integrated Circuits- Analysis T2 & Design, McGraw-Hill, Third Ed., 2003. Jan M. Rabaey, Digital Integrated Circuits- A Design Perspective, Prentice Hall, R3 Second Edition, 2005. R4 Modern VLSI Design – Wayne Wolf, 3 Ed., 1997, Pearson Education. CMOS VLSI Design-A Circuits and Systems Perspective, Neil H.E Weste, R5 David Harris, Ayan Banerjee, 3rd Edn, Pearson, 2009. Online Resources: Sl. No Website Link 1 https://onlinecourses.nptel.ac.in/noc24_ee29/preview 2 https://pages.hmc.edu/harris/cmosvlsi/4e/index.html 3 http://www.vlsi-expert.com/ 4 https://semiwiki.com/ using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain an array multiplier. 4 understanding arithmetic circuits adders: static adder, carry-bypass adder, linear carry- select adder, square- root carry- select adder. multipliers: array multiplier. text /reference: t/r booktitle/author michael john sebastian smith, application specific integrated circuits, pearson t1 education,2001. sung –mo kang & yusuf leblebici, cmos digital integrated circuits- analysis t2 & design, mcgraw-hill, third ed., 2003. jan

m. rabaey, digital integrated circuits- a design perspective, prentice hall, r3 second edition, 2005. r4 modern vlsi design – wayne wolf, 3 ed., 1997, pearson education. cmos vlsi design-a circuits and systems perspective, neil h.e weste, r5 david harris, ayan banerjee, 3rd edn, pearson, 2009. online resources: sl. no website link 1 https://onlinecourses.nptel.ac.in/noc24_ee29/preview 2 https://pages.hmc.edu/harris/cmosvlsi/4e/index.html 3 http://www.vlsi-expert.com/ 4 https://semiwiki.com/ should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain an array multiplier. 4 Understanding Arithmetic circuits Adders: Static adder, Carry-Bypass adder, Linear Carry- Select adder, Square- root carry- select adder. Multipliers: Array multiplier. Text /Reference: T/R BookTitle/Author Michael John Sebastian Smith, Application Specific Integrated Circuits, Pearson T1 Education,2001. Sung –Mo Kang & Yusuf Leblebici, CMOS Digital Integrated Circuits-Analysis T2 & Design, McGraw-Hill, Third Ed., 2003. Jan M. Rabaey, Digital Integrated Circuits- A Design Perspective, Prentice Hall, R3 Second Edition, 2005. R4 Modern VLSI Design – Wayne Wolf, 3 Ed., 1997, Pearson Education. CMOS VLSI Design-A Circuits and Systems Perspective, Neil H.E Weste, R5 David Harris, Ayan Banerjee, 3rd Edn, Pearson, 2009. Online Resources: Sl. No Website Link 1 https://onlinecourses.nptel.ac.in/noc24_ee29/preview 2 https://pages.hmc.edu/harris/cmosvlsi/4e/index.html 3 http://www.vlsi-expert.com/ 4 https://semiwiki.com/ means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Explain an array multiplier. 4 Understanding Arithmetic circuits Adders: Static adder, Carry-Bypass adder, Linear Carry- Select adder, Square- root carry- select adder. Multipliers: Array multiplier. Text /Reference: T/R BookTitle/Author Michael John Sebastian Smith, Application Specific Integrated Circuits, Pearson T1 Education,2001. Sung –Mo Kang & Yusuf Leblebici, CMOS Digital Integrated Circuits- Analysis T2 & Design, McGraw-Hill, Third Ed., 2003. Jan M. Rabaey, Digital Integrated Circuits- A Design Perspective, Prentice Hall, R3 Second Edition, 2005. R4 Modern VLSI Design – Wayne Wolf, 3 Ed., 1997, Pearson Education. CMOS VLSI Design-A Circuits and Systems Perspective, Neil H.E Weste, R5 David Harris, Ayan Banerjee, 3rd Edn, Pearson, 2009. Online Resources: Sl. No Website Link 1 https://onlinecourses.nptel.ac.in/noc24_ee29/preview 2 https://pages.hmc.edu/harris/cmosvlsi/4e/index.html 3 http://www.vlsi-expert.com/ 4 https://semiwiki.com/ □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

▶ **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

```
Definition → Principle →
Components/steps →
Application → Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

```
Aim → Apparatus → Procedure
→ Observation → Result →
Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

```
The official assessment structure is reproduced from the supplied PDF.
```

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

▶ **Define or explain 3 Understanding ASICs. (3 marks)**

▶ **Define or explain Explain the processes involved in SoC design.. (3 marks)**

▶ **Define or explain Outline the FPGA design flow 4 Understanding Compare the speed, power and area. (3 marks)**

▶ **Define or explain 6 Understanding considerations in VLSI VLSI Design Methodologies. Introduction: Moore's law .ASIC design, Full custom ASICs, Standard cell based ASICs, Gate array based ASICs, SoCs, FPGA devices, FPGA Design flows, Top-Down and Bottom-Up design methodologies. Speed power and area considerations in VLSI design. (3 marks)**

Module 2

▶ **Define or explain 3 Understanding inverter Realize the given logic function using static. (3 marks)**

▶ **Define or explain 3 Applying CMOS logic and transmission gate logic. Compare and contrast static and dynamic. (3 marks)**

▶ **Define or explain 2 Understanding CMOS logic. Realize the given logic**

function using pass. (3 marks)

▶ **Define or explain 4 Applying transistor logic and transmission gate logic. Static CMOS Logic Design MOSFET Logic Design - NMOS Inverter (Static analysis only), basic logic gates, CMOS logic, Static and transient analysis of CMOS inverter. Realization of logic functions with static CMOS logic, Pass transistor logic, and transmission gate logic. (3 marks)**

Module 3

▶ **Define or explain 3 Understanding and its drawbacks.. (3 marks)**

▶ **Define or explain Explain NP domino logic.. (3 marks)**

▶ **Define or explain Realize OR, NOR and NAND ROM cells.. (3 marks)**

▶ **Define or explain Compare and contrast SRAM and DRAM 2 Understanding Dynamic logic Design and Storage Cells Dynamic Logic Design-Pre charge- Evaluate logic, Domino Logic, NP domino logic. Read Only Memory-4x4 MOS ROM Cell Arrays(OR,NOR,NAND) Random Access Memory –SRAM-Six transistor CMOS SRAM cell, DRAM –Three transistor and One transistor Dynamic Memory Cell... (3 marks)**

Module 4

▶ **Define or explain 2 Understanding bypass adder.. (3 marks)**

▶ **Define or explain Explain Linear carry select adder working. 5 Understanding Explain the working of square root carry. (3 marks)**

▶ **Define or explain 5 Understanding select adder.. (3 marks)**

▶ Define or explain Explain an array multiplier. 4 Understanding Arithmetic circuits Adders: Static adder, Carry-Bypass adder, Linear Carry- Select adder, Square- root carry- select adder. Multipliers: Array multiplier. Text /Reference: T/R BookTitle/Author Michael John Sebastian Smith, Application Specific Integrated Circuits, Pearson T1 Education,2001. Sung –Mo Kang & Yusuf Leblebici, CMOS Digital Integrated Circuits- Analysis T2 & Design, McGraw-Hill, Third Ed., 2003. Jan M. Rabaey, Digital Integrated Circuits- A Design Perspective, Prentice Hall, R3 Second Edition, 2005. R4 Modern VLSI Design – Wayne Wolf, 3 Ed., 1997, Pearson Education. CMOS VLSI Design-A Circuits and Systems Perspective, Neil H.E Weste, R5 David Harris, Ayan Banerjee, 3rd Edn, Pearson, 2009. Online Resources: Sl. No Website Link 1 https://onlinecourses.nptel.ac.in/noc24_ee29/preview 2 https://pages.hmc.edu/harris/cmosvlsi/4e/index.html 3 http://www.vlsi-expert.com/ 4 https://semiwiki.com/. (3 marks)

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **3 Understanding ASICs**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **3 Understanding inverter Realize the given logic function using static**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **3 Understanding and its drawbacks.**.

3-mark explanation: Explain one principle, process or comparison from this module.

Module 4

1-mark definition: State the meaning of **2 Understanding bypass adder.**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl. No Website Link
- https://onlinecourses.nptel.ac.in/noc24_ee29/preview
- https://pages.hmc.edu/harris/cmosvlsi/4e/index.html http://www.vlsi-expert.com/ https://semiwiki.com/

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